FYS4411/9411 March 20

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mean (sample) value

$$\bar{x} = \mu_x = \frac{1}{n} \sum_{i=1}^{n} x_i$$

variance of $\bar{x}$

$$\mathrm{var}[\bar{x}] = \sigma^2 / n$$

$$\sigma^2 = \mathrm{var}[x]$$

$$\sigma_n^2 = \mathrm{var}[\bar{x}] =$$

$$\frac{1}{n^2} \sum_{i=1}^{n} \sum_{j=1}^{n} \underbrace{(x_i - \mu)(x_j - \mu)}_{\mathrm{cov}(x_i, x_j)}$$

$\mathrm{cov}(x_i, x_j) \neq 0$ when x_i and x_j are not independent

$$\underbrace{\frac{1}{n^2} \sum_{i=1}^{n} (x_i - \mu)^2}_{\sigma^2/n} +$$

$$\frac{2}{n} \sum_{k<l} (x_k - \mu)(x_l - \mu)$$

$$= \frac{1}{n^2} \sum_{i=1}^{n} \sum_{j=1}^{n} \operatorname{cov}(x_i, x_j)$$

$$= \frac{1}{n^2} \sum_{i=1}^{n} \sum_{j=1}^{n} \Sigma_{ij}$$

Rewrite the $\sigma^2/n + \frac{2}{n} \Sigma \ldots$

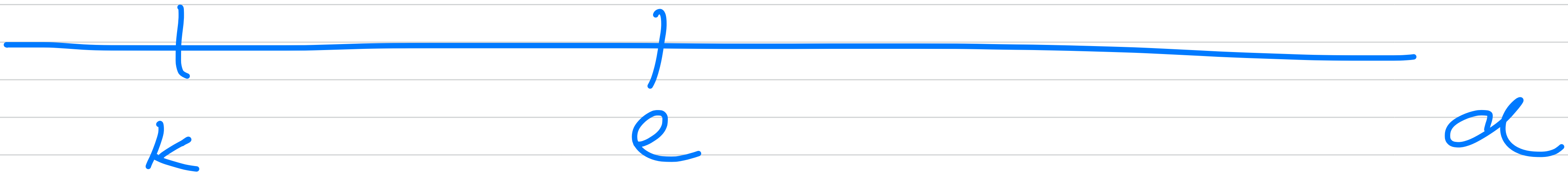
introduce a function

fd

$$\rho_d = \frac{1}{n-d}\sum_{k=1}^{n-d}(x_k-\mu)(x_{k+d}-\mu)$$

$$\ell = k + d \longrightarrow$$

$$d = |\ell - k|$$



$$K_d = \frac{f_d}{\sigma^2}$$

$$d = 0 \Rightarrow f_0 = \sigma^2$$

$$\sigma_n^2 = \frac{\sigma^2}{n} + \frac{2}{n} \sum_{d=1}^{n-1} f_d$$

$$= \left(1 + 2 \sum_{d=1}^{n-1} K_d\right) \frac{\sigma^2}{n}$$

autocorrelation time τ

$$\tau = 1 + 2 \sum_{d=1}^{n-1} \kappa_d$$

$$\sigma_n^2 = \frac{\tau}{n} \sigma^2$$

interpretation: τ represents the number of measurements needed in order to establish when we can consider an experiment to consist of independent events

$\mathrm{cov}(x_i, x_j)$ depends only on the distance $|i-j|$

$$\Sigma_{ij} = \mathrm{cov}(|i-j|) = c(|i-j|)$$

$$\sigma_n^2 = \frac{1}{n^2} \sum_i \sum_j c(|i-j|)$$

In Flyvbjerg & Petersen

$$t = |i-j|$$

for $t = 0$, then $\sigma_n^2 = \sigma^2/n$

we have then n-terms

for $t \geq 1$, there are $2(n-t)$

terms since both (i,j)

and (j,i) contribute

$$c(|i-j|) = c(t)$$

$$\sigma_n^2 = \frac{1}{n^2}\Big(n\underbrace{c(0)}_{\sigma^2} + 2\sum_{t=1}^{n-1}(n-t)c(t)\Big)$$

$$K_d \to K(t) = \frac{C(t)}{C(0)}$$

$$= \frac{C(t)}{\sigma^2}$$

$$\sigma_n^2 = \frac{\sigma^2}{n}\left(1 + \frac{2}{n}\sum_{t=1}^{n-1}(n-t)K(t)\right)$$

$$= \frac{\sigma^2}{n}\left(1 + 2\sum_{t=1}^{n-1}\left(1 - \frac{t}{n}\right)K(t)\right)$$

$$(n \to \infty)$$

$$\sum_{t=1}^{\infty} |k(t)| < \infty$$

for large n $\quad 1 - \frac{t}{n} \simeq 1$

$$\sigma_n^2 \simeq \frac{\sigma^2}{n}\left(1 + 2\sum_{t=1}^{n-1} k(t)\right)$$

Idea behind blocking

Don't evaluate $cov(x_i, x_j)$

- combine neighboring samples into blocks
- ~~Each~~ Each blocking level reduce correlations if Blocks are large enough $\simeq$ independent samples

Start with dataset

$x_i^{(0)}$ $i = 1, 2, \ldots n$

Define the blocking transformation as

$$x_i^{(k+1)} = \frac{1}{2}\left(x_{2i-1}^{(k)} + x_{2i}^{(k)}\right)$$

$$\left(2^m\right)$$

Properties

- number of samples halves each iteration

- correlations decrease with blocking level

after k-transformations

$$N_k = \frac{n}{2^k}$$

compute

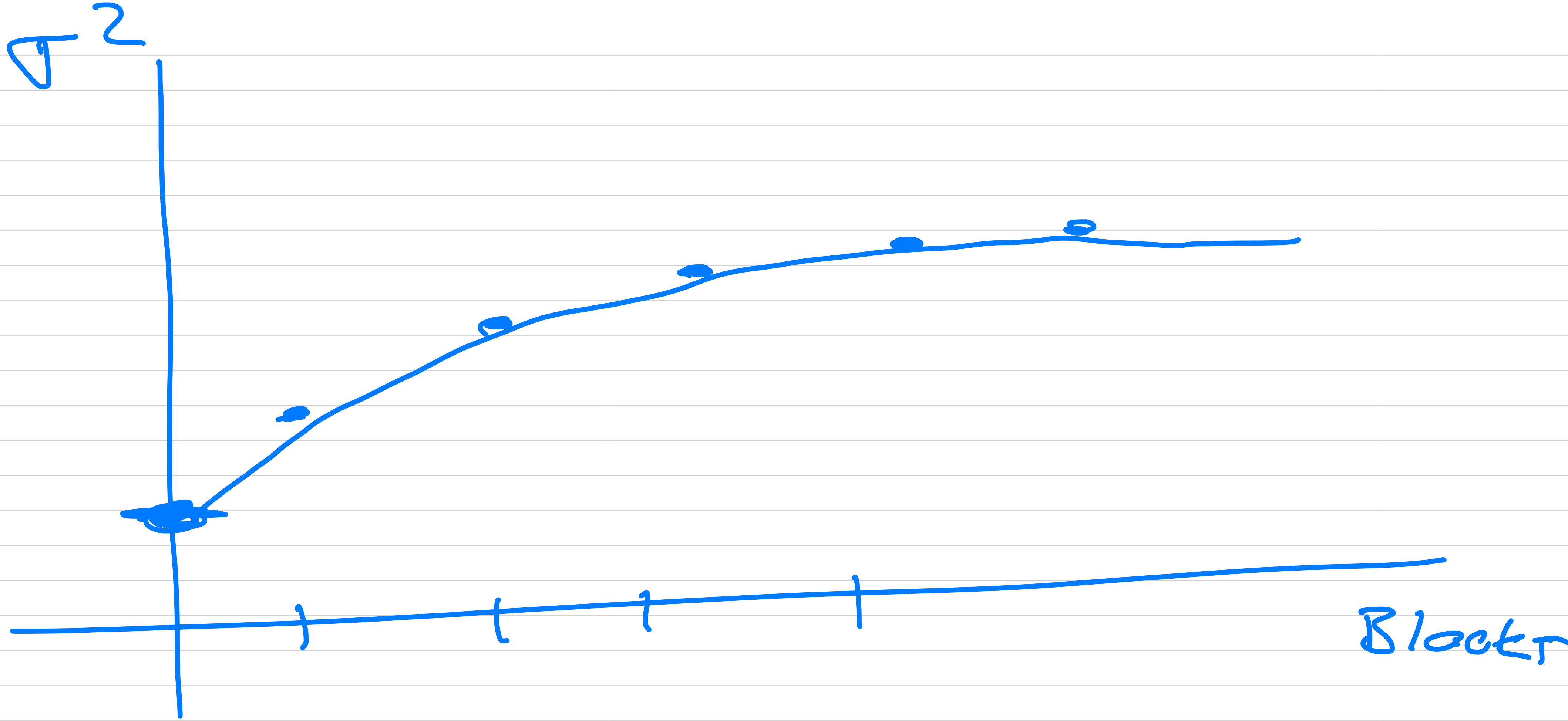
$$\bar{X}_k = \frac{1}{N_k} \sum_{i=1}^{N_k} x_i^{(k)}$$

estimate variance

$$s_k^2 = \frac{1}{N_k - 1} \sum_{i=1}^{N_k} \left(x_i^{(k)} - \bar{X}_k \right)$$

Error estimate

$$\sigma_k^2 = s_k^2 / N_k \,, \quad \sigma_k^2 \to \sigma^2 / n$$



$$\sigma_0^2 = \sigma^2/n$$

$$\sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2$$

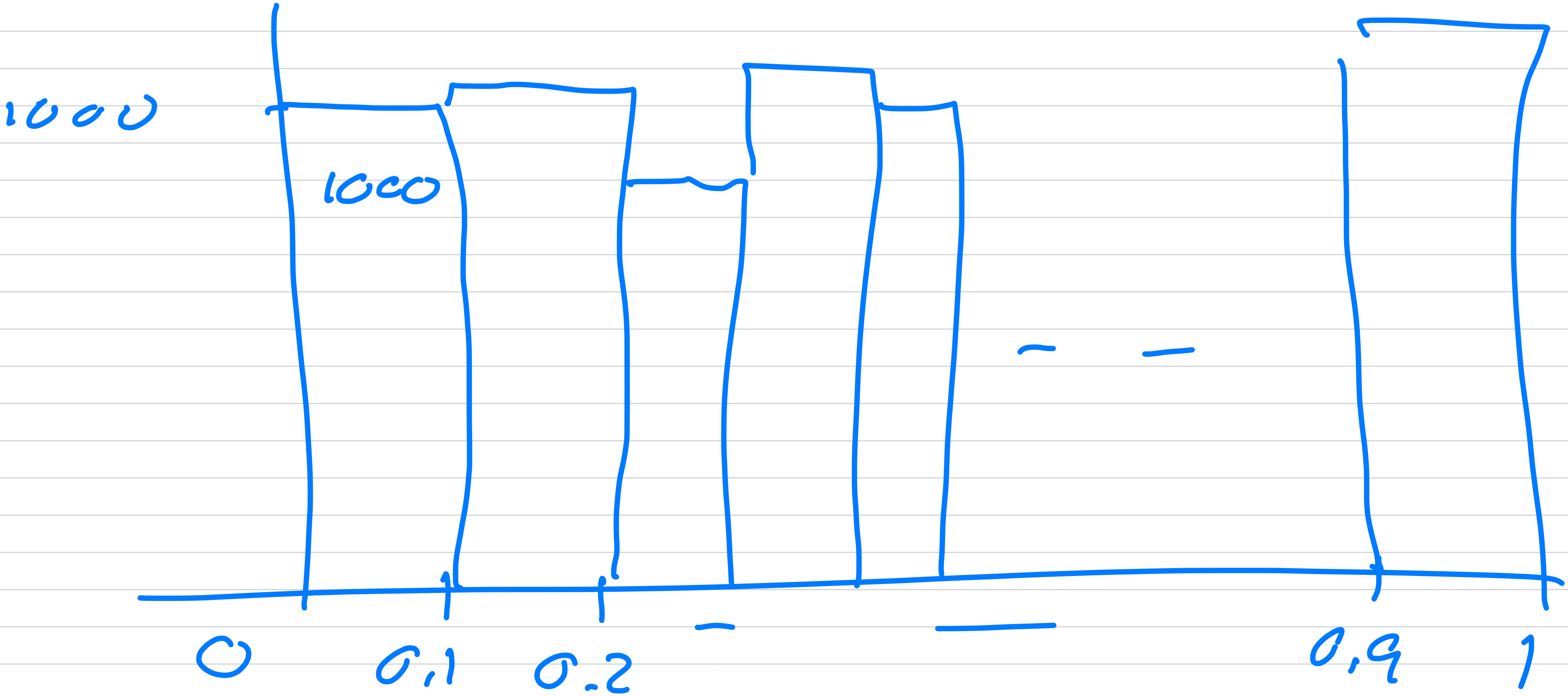
$$p(x)\,dx = \begin{cases} 1 & 0 \leq x \leq 1 \\ 0 & \text{else} \end{cases}$$

$$\bar{x} = \langle x \rangle = \mu = \int_0^1 x\,p(x)\,dx$$

$$= 1/2$$

$$\sigma^2 = \int_0^1 (x-\mu)^2 p(x)\,dx = \frac{1}{12}$$

$$\text{STD} = \frac{1}{\sqrt{12}}$$



N = 10000